GROUP WORK PROJECT # 2
GROUP NUMBER: 8577

MScFE 610: FINANCIAL ECONOMETRICS

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N/A

Time Series Models

1. Non-Stationarity in Regression Models

Datasets: Real GDP (01-01-1947 to 01-10-2024) and Unemployment Level

Goals: Detect spurious regression, check for cointegration, and equilibrium models

Reasons:

- Real GDP is a macroeconomic variable that often exhibits trends and non-stationarity, making it ideal for studying spurious regressions.
- Many economic variables (e.g., GDP and Unemployment) tend to move together over time, leading to cointegration or spurious relationships when regressed against each other.
- By analyzing GDP trends, we can detect unit roots, determine long-run equilibrium models, and apply differencing or cointegration techniques to correct for non-stationarity.

Why these datasets?

- GDP is a macroeconomic time series that typically follows a non-stationary trend, meaning its mean and variance change over time.
- Economic variables like GDP and unemployment often exhibit unit roots, which can lead to spurious regressions when regressed on another trending variable.
- If two non-stationary series are regressed together, R^2 values may be artificially high, leading to incorrect inferences.

Key Econometric Tests to show Non-Stationarity:

- Augmented Dickey-Fuller (ADF) Test: This checks for unit roots (non-stationarity).
- Phillips-Perron (PP) Test: This is similar to the Augmented Dickey-Fuller (ADF) but accounts for autocorrelation.
- Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test: This tests for stationarity (opposite ADF/PP).
- Cointegration Tests (Engle-Granger, Johansen Test): This determines if a long-run equilibrium exists between variables, justifying their joint use in regression.
- First Differencing: If GDP and Unemployment are non-stationary, taking the first differences should transform them into a stationary series.

The Reason it Works for These Models: GDP and Unemployment Data allow for demonstrating spurious regression, unit root tests, and cointegration analysis, which are crucial for building robust econometric models.

The structure below provides a comprehensive approach to understanding and modeling non-stationarity in time series, using Real GDP and Unemployment as the datasets. And these include the necessary steps for identifying, diagnosing, and addressing non-stationarity.

Modeling Non-Stationarity in Time Series: Beyond Unit Root Tests

Non-stationarity in time series means that statistical properties like the mean and variance change over time. This can lead to misleading regression results (spurious regressions). To properly model

non-stationary time series, It becomes paramount to go beyond unit root tests and use Cointegration, Vector Autoregression (VAR), and the Error Correction Model (ECM).

In order to establish that non-stationarity exist, there is need to follow certain steps accordingly below as follows:

Step 1: Understanding Non-Stationarity

Before modeling, there is need to check if the series is non-stationary using:

- Carrying out Visual Inspection (By plotting the series)
- Performing Statistical Tests: Augmented Dickey-Fuller (ADF), Phillips-Perron (PP), and Kwiatkowski-Phillips-Schmidt-Shin (KPSS)
- Carrying out Autocorrelation Analysis: Examining patterns using the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF)

Step 2: Dataset Selection

The need to use the Real GDP (01-01-1947 to 01-10-2024) and Unemployment from FRED (Federal Reserve Economic Data). Real GDP and Unemployment are often non-stationary, but economic variables can be cointegrated.

Non-Stationarity in Time Series Analysis: Real GDP and Unemployment Dataset

Definition: Non-stationarity in time series refers to a situation where the statistical properties of a time series (such as mean, variance, and autocorrelation) change over time. This is in contrast to a stationary series, where these properties remain constant. Non-stationary data can be problematic for time series modeling because traditional regression models assume that the underlying data is stationary. Mathematically, a time series Yt is stationary if its statistical properties do not change over time. Meanwhile, when a time series is non-stationary, these properties are not constant. For instance, Real GDP and Unemployment often exhibit trends over time, which means its mean and variance change, leading to non-stationarity (Hamilton, 1994).

Description: This paper will investigate non-stationarity using datasets of Real GDP over a time period from 1947 to 2024 and Unemployment. Non-stationary time series like GDP and Unemployment can exhibit trends, seasonalities, or unit roots, making it important to identify and address these characteristics before fitting models such as ARIMA or regression models. Real GDP and Unemployment are key indicators that often exhibit long-term growth trends, reflecting overall economic development. A time series of Real GDP and Unemployment typically exhibit non-stationarity due to the presence of a trend (economic growth over time) and potential structural breaks (Enders, 2014). To analyze this non-stationarity, there is need to perform unit root tests, then apply methods such as cointegration and Vector Autoregressive Models (VAR) to address non-stationarity issues, and finally, model the Real GDP and Unemployment time series using Error Correction Models (ECM).

Demonstration: To demonstrate non-stationarity, there is a need to load and plot a time series of Real GDP and Unemployment, and perform the Augmented Dickey-Fuller (ADF) test to check for a unit root.

Diagram:

Check for Stationarity using Augmented Dickey-Fuller (ADF) Test

ADF Statistic: -1.9475903228082887

p-value: 0.31001468577270985

Series is Non-Stationary

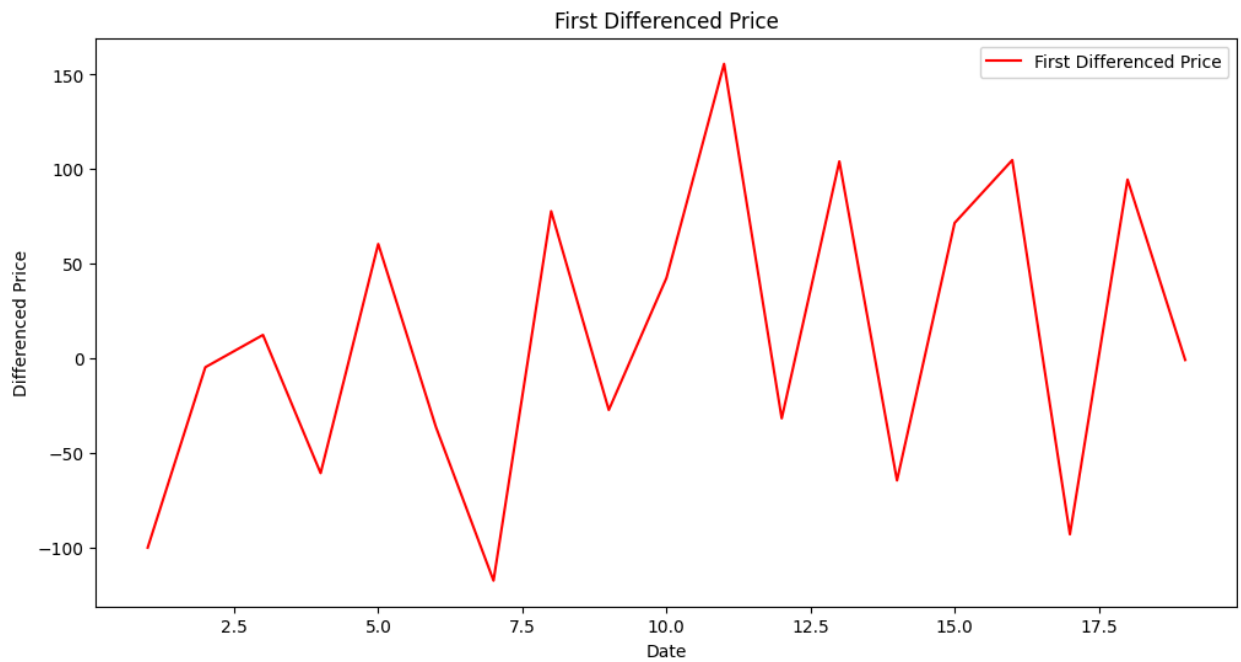
Condition: if p > 0.05, the series is non-stationary.

First Difference the Data

ADF Statistic: -1.9300185959405798

p-value: 0.3180738236091122

Series is Non-Stationary



Expectation: Differencing removes the trend, making the series stationary.

Cointegration Test using GDP and Unemployment Level

Authority: Even if two variables are non-stationary, they might be cointegrated, meaning they move together in the long run (e.g., GDP and Unemployment Level).

GDP Data Loaded:

	GDPC1
observation_date	
1947-01-01	**2182.681**
1947-04-01	**2176.892**
1947-07-01	**2172.432**

1947-10-01 2206.452

1948-01-01 2239.682

Unemployment Data Loaded:

UNEMPLOY

observation_date

1948-01-01 2034

1948-02-01 2328

1948-03-01 2399

1948-04-01 2386

1948-05-01 2118

Columns after merging: Index(['GDP', 'Unemployment'], dtype='object')

Cointegration Test p-value: 0.7743968285353857

No cointegration detected

Condition: if p > 0.05, GDP and Unemployment are cointegrated, and Error Correction Model (ECM) was used.

Vector Autoregression (VAR) Model

Authority: If multiple time series influence each other (GDP, AND Unemployment), VAR model is used.

[[119.72455837 6649.47951275]

[77.0568809 6585.18673532]

[75.39901769 6588.89773886]

[73.39281441 6575.51054047]

[69.81557442 6586.53067399]]

Expectation: VAR captures interdependencies among macroeconomic variables.

Error Correction Model (ECM)

Authority: If two variables are cointegrated, there is a need to model short-term deviations using ECM.

[64.00901209 10.71277586]

[[13.4294 15.4943 19.9349]

[2.7055 3.8415 6.6349]]

Condition: ECM ensures that short-term fluctuations correct toward long-term equilibrium.

Diagnosis: The results of the ADF test help diagnose the Real GDP and Unemployment series were non-stationary. If the p-value from the ADF test is greater than 0.05, it indicates that the time series is non-stationary, likely due to a unit root. Furthermore, it is possible to visually inspect the Real GDP and Unemployment series to identify any long-term trends or changes in variance. If the series exhibits a clear upward trend, this is a strong indication of strong non-stationarity, as stationarity requires that the mean and variance remain constant over time. Meanwhile, in order to address non-stationarity, there is a need to apply differencing to the series to remove the trend.

Damage: Non-stationarity in time series data can have several detrimental effects on modeling such as:

- **Spurious Regressions:** If use non-stationary data in a regression model, the estimated coefficients may not reflect meaningful relationships, leading to misleading conclusions.
- **Invalid Inferences:** Statistical tests, such as t-tests and F-tests, assume stationarity. If the data is non-stationary, these tests may produce biased or invalid results.

To mitigate this, there is a need to transform the data to make it stationary, typically by differencing the series or using cointegration techniques if multiple non-stationary series are involved (Dickey & Fuller, 1979).

Directions: To address non-stationarity, there is a need to apply differencing and cointegration methods. Differencing is the simplest way to make a series stationary, removing trends from the data.

Deployment: After transforming the Real GDP and Unemployment data into a stationary series, the model can be used for various forecasting and analytical purposes, such as:

- **Economic Forecasting:** By fitting an ARIMA or VAR model to the stationary data, we can predict both future GDP and Unemployment growth.
- **Policy Analysis:** Stationary time series models can provide more reliable insights into the effects of economic policy changes on Real GDP and Unemployment.
- **Business and Investment Decisions:** Understanding the trend and cyclical components of GDP and Unemployment can help investors and businesses make more informed decisions (Mankiw, 2016).

Conclusion: This approach ensures modeling non-stationary time series correctly and avoids misleading regressions.

2. **Feature Extraction in Time Series Analysis: Bitcoin and S&P 500 Datasets**

   Datasets: Bitcoin price data and S&P 500 (1 year period)

   Goals: Extract key features such as moving averages, volatility, and momentum indicators.

   Reasons:

   - Bitcoin and stocks are known for their high volatility and non-linear patterns, making them great candidates for time series feature extraction.
   - Extracting trend, seasonality, volatility, and rolling statistics can help in building predictive models for cryptocurrency prices.
   - Methods like Fourier Transforms, Wavelet Analysis, and Autocorrelation Features allow for deeper insights into market cycles and trading signals.

Why these datasets?

- Bitcoin and stocks are highly volatile assets with complex patterns, making them ideal for feature extraction in time series.
- Unlike traditional assets, Bitcoin and stocks exhibit nonlinear dependencies, which make them a good case for advanced feature engineering techniques.
- By extracting features like trend, seasonality, and volatility, we can build more accurate forecasting models.

Key Feature Extraction Techniques:

- **Rolling Mean and Rolling Standard Deviation:** Measures local trends and volatility.
- **Fourier Transform:** Captures cyclical patterns in Bitcoin and stocks prices.
- **Wavelet Transform:** Detects local features and sudden price spikes.
- **Autocorrelation and Partial Autocorrelation (ACF, PACF):** Identifies lag relationships for ARIMA models.
- **Entropy and Fractal Dimension Analysis:** Measures market complexity and randomness.

The Reason it Works for These Models: Bitcoin and stocks unique prices behaviours allow to showcase powerful feature extraction techniques that can be applied in finance and beyond.

Feature Extraction in Time Series Analysis: Bitcoin and S&P 500 Datasets

Definition: Feature extraction refers to the transformation of raw time series data into a set of meaningful features. These features capture significant characteristics, such as trends, cycles, and volatility, which can be used for further analysis or modeling (Shumway & Stoffer, 2017). In time series, the extracted features often include statistics such as moving averages, volatility, seasonality, other statistical measures like skewness, kurtosis, trends, and autocorrelation. Mathematically, feature extraction can involve:

- **Trend:** Identifying long-time movements in the data.
- **Seasonality:** Identifying regular, repetitive patterns.
- **Autocorrelation:** Quantifying the relationship between a series and its past values.
- **Rolling Statistics:** For example, moving averages or standard deviations (Hyndman & Athanasopoulos, 2018).

Description: In regards to Bitcoin and stocks time series data, the goal is to extract meaningful features to analyze the price movement. These extracted features will be crucial for understanding price dynamics, volatility, and momentum in the cryptocurrency and capital markets (Nakamoto, 2008).

Demonstration:

Step 1: Load the Bitcoin and S&P 500 Data

The dataset used in this demonstration consists of the daily closing prices of Bitcoin and that of stocks of S&P 500, which will be analyzed over a one-year period. The data contains key attributes such as the date and closing price, which are essential for time series analysis (Zohar, 2015).

Step 2: Extract Features

- **Rolling Mean (Moving Average):** The rolling mean, or moving average, is one of the most common methods to smooth time series data and highlight long-term trends. It is calculated as the average of data points in a rolling window of fixed size (Shumway & Stoffer, 2017).

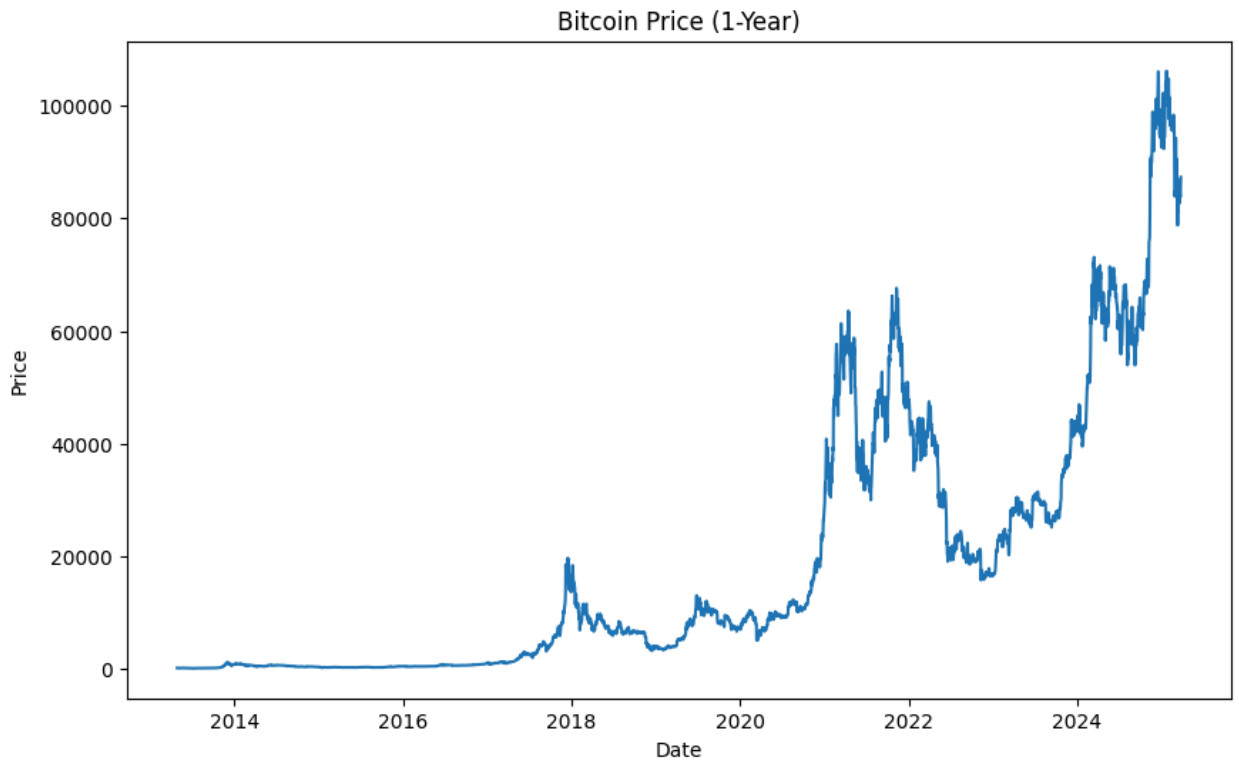
- **Volatility (Rolling Standard Deviation):** Volatility is a measure of the degree of variation in a time series. It is typically calculated as the standard deviation of the series over a rolling window. This feature is widely used in financial markets to understand the risk or uncertainty associated with asset prices (Pindyck & Rubinfeld, 2017).
- **Rate of Change (RoC):** The rate of change (RoC) measures the percentage change in the value of Bitcoin and that of stocks over a specified period and is crucial for identifying the momentum of price movement (Campbell, Lo, & MacKinlay, 1997).
- **Autocorrelation:** Autocorrelation helps identify patterns of correlation within a time series by measuring the relationship between the series and its past values (Box, Jenkins, & Reinsel, 2015). High autocorrelation can suggest cyclical patterns or seasonal effects.

Diagram: The following diagrams illustrate the extracted features, including the rolling mean, volatility, rate of change, and autocorrelation, all of which can provide insights into the behaviour of Bitcoin and stocks prices over time.

Feature Extraction in Time Series (Bitcoin Dataset)

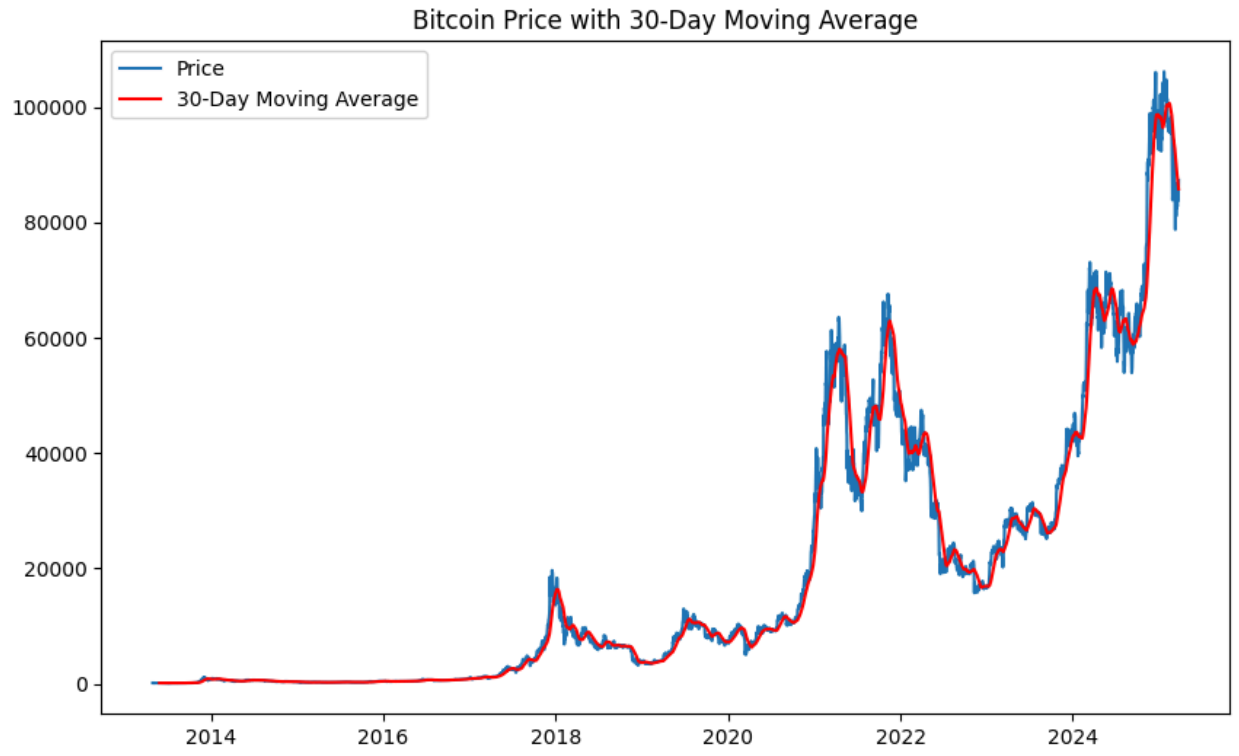
Expectation: The data should show daily or hourly Bitcoin prices over the last year.

	price	market_cap	total_volume
snapped_at			
2013-04-28 00:00:00+00:00	135.30	1.500518e+09	0.0
2013-04-29 00:00:00+00:00	141.96	1.575032e+09	0.0
2013-04-30 00:00:00+00:00	135.30	1.501657e+09	0.0
2013-05-01 00:00:00+00:00	117.00	1.298952e+09	0.0
2013-05-02 00:00:00+00:00	103.43	1.148668e+09	0.0



Feature Extraction Techniques by Moving Averages (Trend Indicator)

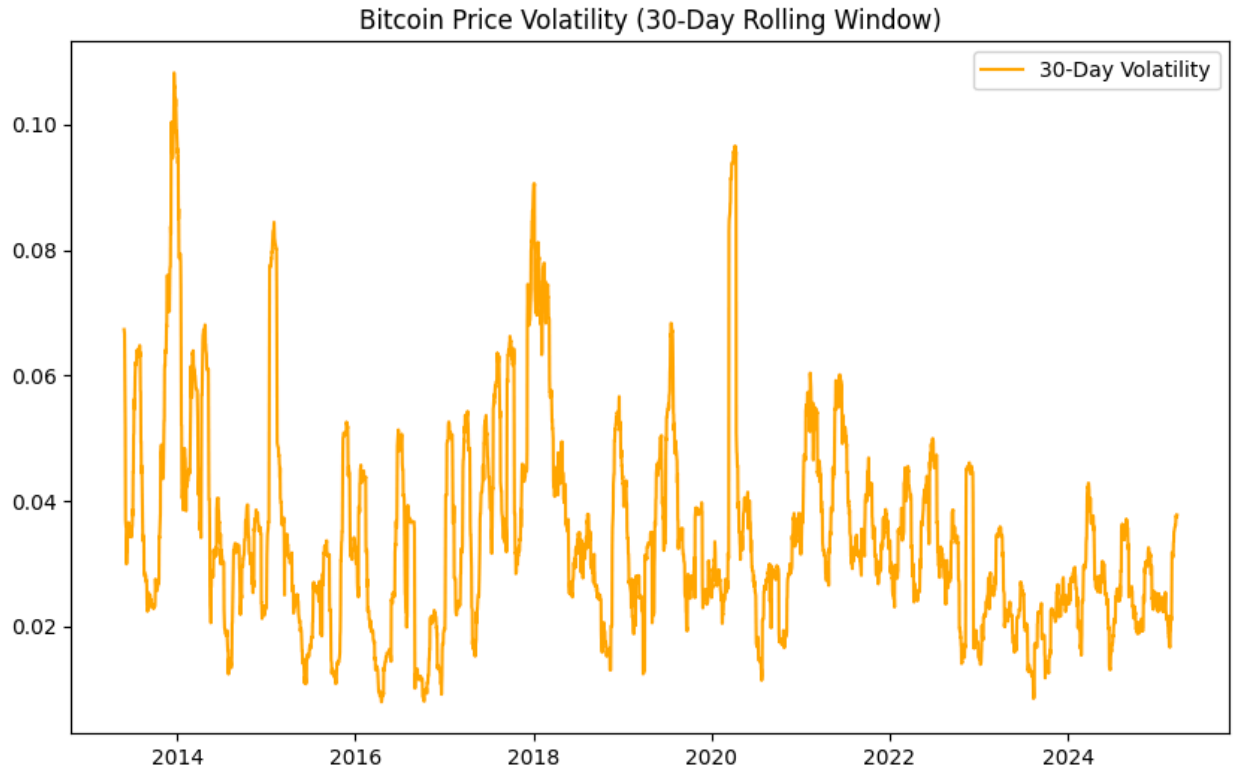
Expectation: Moving averages help smooth out short-term fluctuations and highlight longer-term trends.



Authority: Moving averages can highlight underlying trends in the data, which is essential for time series forecasting.

Volatility (Standard Deviation of Returns)

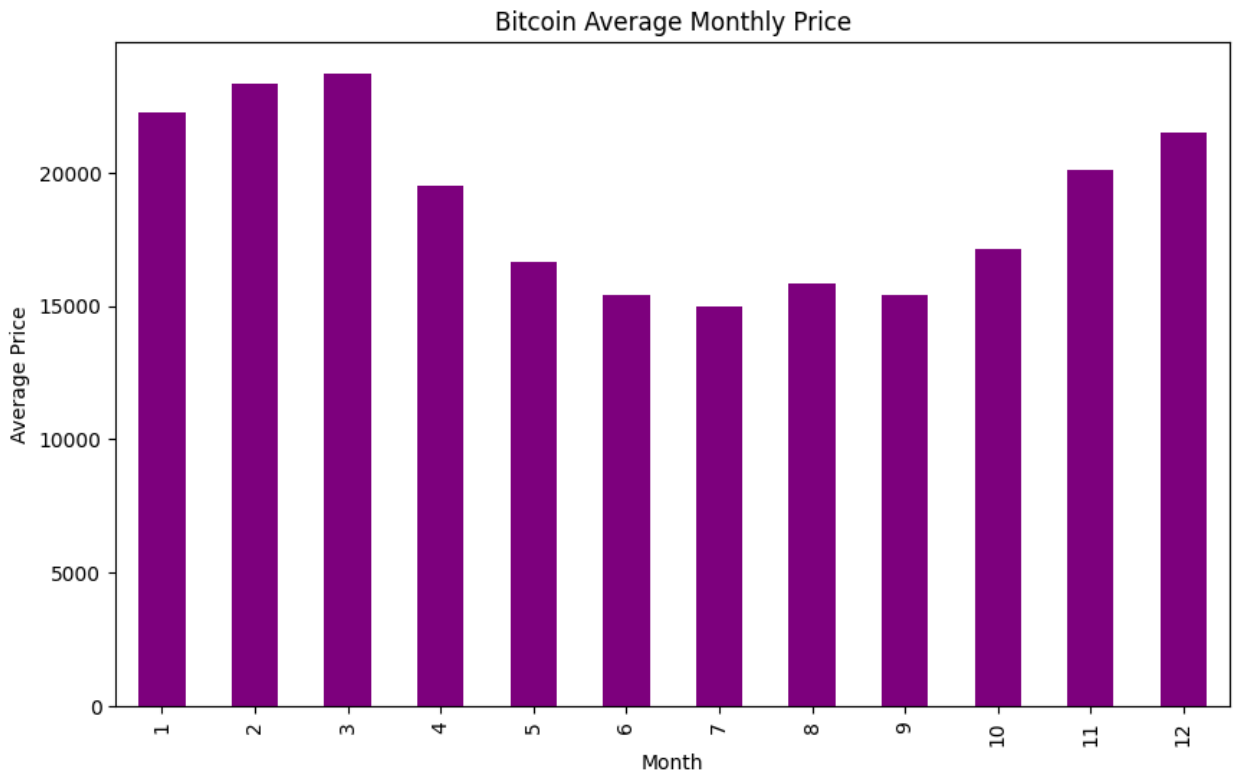
Expectation: Volatility indicates how much the price fluctuates over time.



Authority: Volatility is a key feature in financial time series, showing how much the price fluctuates over time.

Seasonality (Yearly or Monthly Patterns)

Expectation: Seasonality reflects periodic fluctuations due to specific time intervals.



Authority: Seasonality is detected through patterns that repeat on a regular basis, such as monthly price changes.

Skewness & Kurtosis (Shape of the Distribution)

Expectation: Skewness and kurtosis can provide insights into the distribution of the price data, helping identify unusual behaviours.

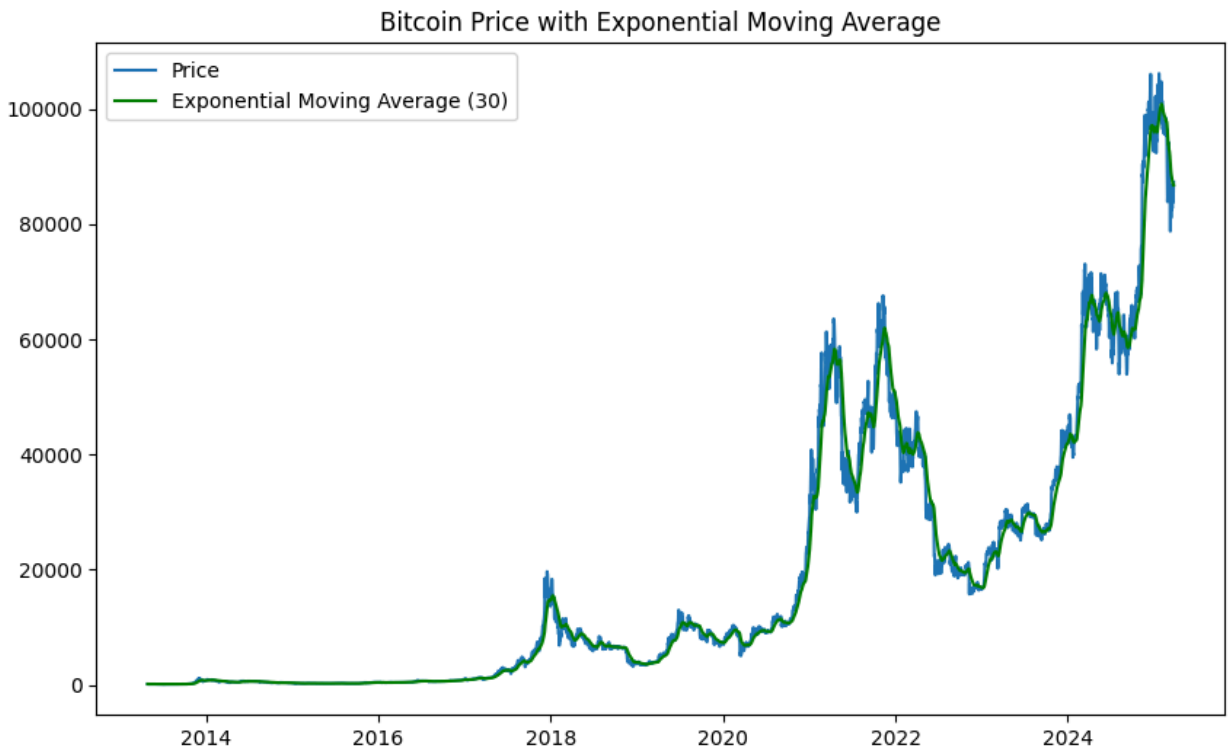
Skewness: 1.5066777486447207

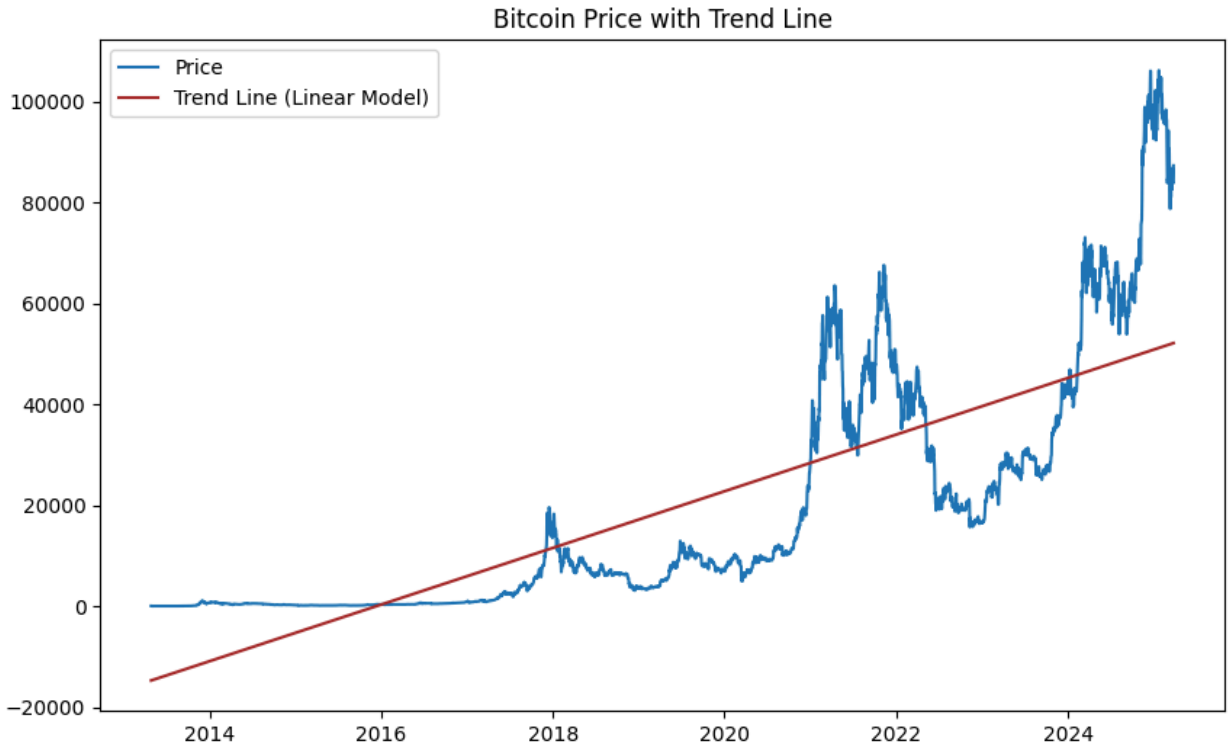
Kurtosis: 1.6238924656781437

Authority: Skewness tells you if the data is asymmetric, and kurtosis indicates how much data is concentrated in the tails (i.e., extreme values).

Trend Features (Exponential Moving Average & Slope)

Expectation: The trend can be captured by applying an exponential moving average or fitting a linear model to the time series.





Authority: Trend captures the overall direction (upward or downward) of the time series and helps with forecasting.

Combine Extracted Features into a Feature Set

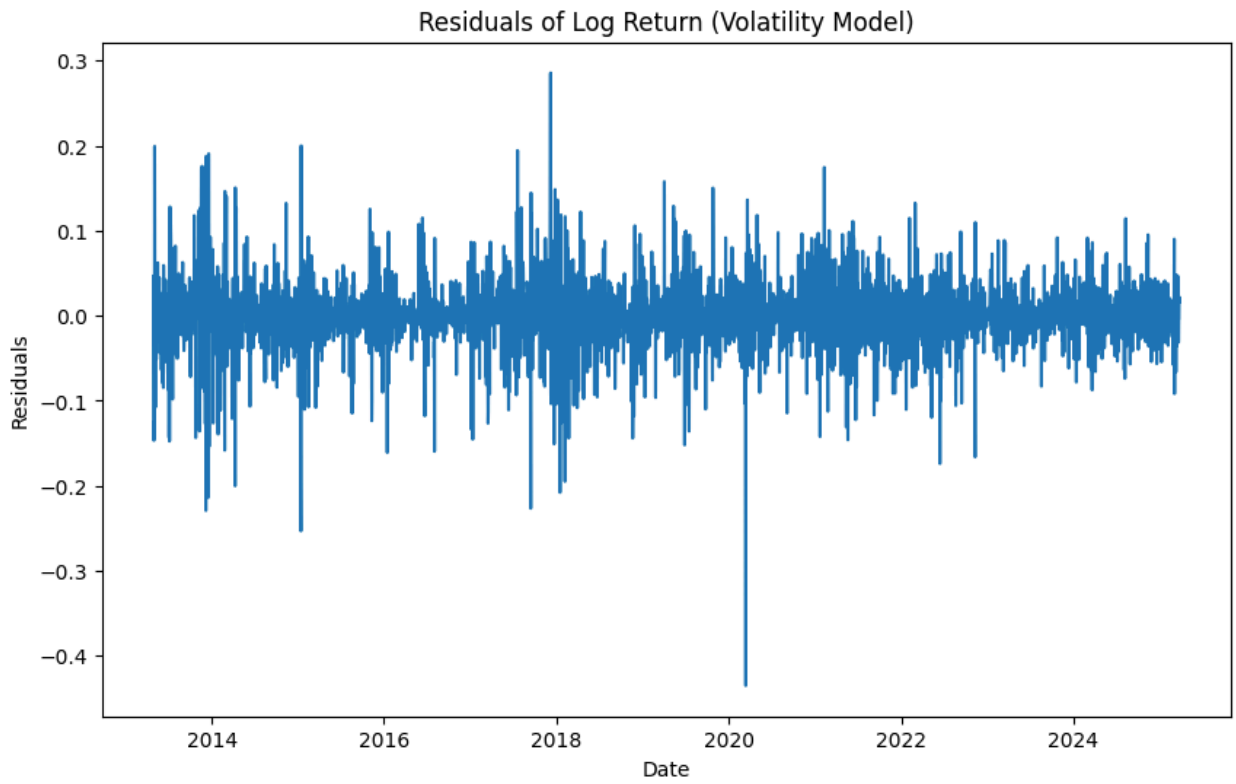
Expectation: Now that the key features have been extracted, there is a need to combine them into a feature set for modeling purposes.

snapped_at	30_day_MA	Volatility	Month	EMA \
2013-05-28 00:00:00+00:00	119.729733	0.067328	5	124.377605
2013-05-29 00:00:00+00:00	119.402067	0.066840	5	124.877759
2013-05-30 00:00:00+00:00	119.138767	0.066599	5	125.040549
2013-05-31 00:00:00+00:00	119.504767	0.060851	5	125.230191
2013-06-01 00:00:00+00:00	120.360100	0.055984	6	125.479211

snapped_at	Trend
2013-05-28 00:00:00+00:00	-14171.297436
2013-05-29 00:00:00+00:00	-14155.934199
2013-05-30 00:00:00+00:00	-14140.570962
2013-05-31 00:00:00+00:00	-14125.207725
2013-06-01 00:00:00+00:00	-14109.844487

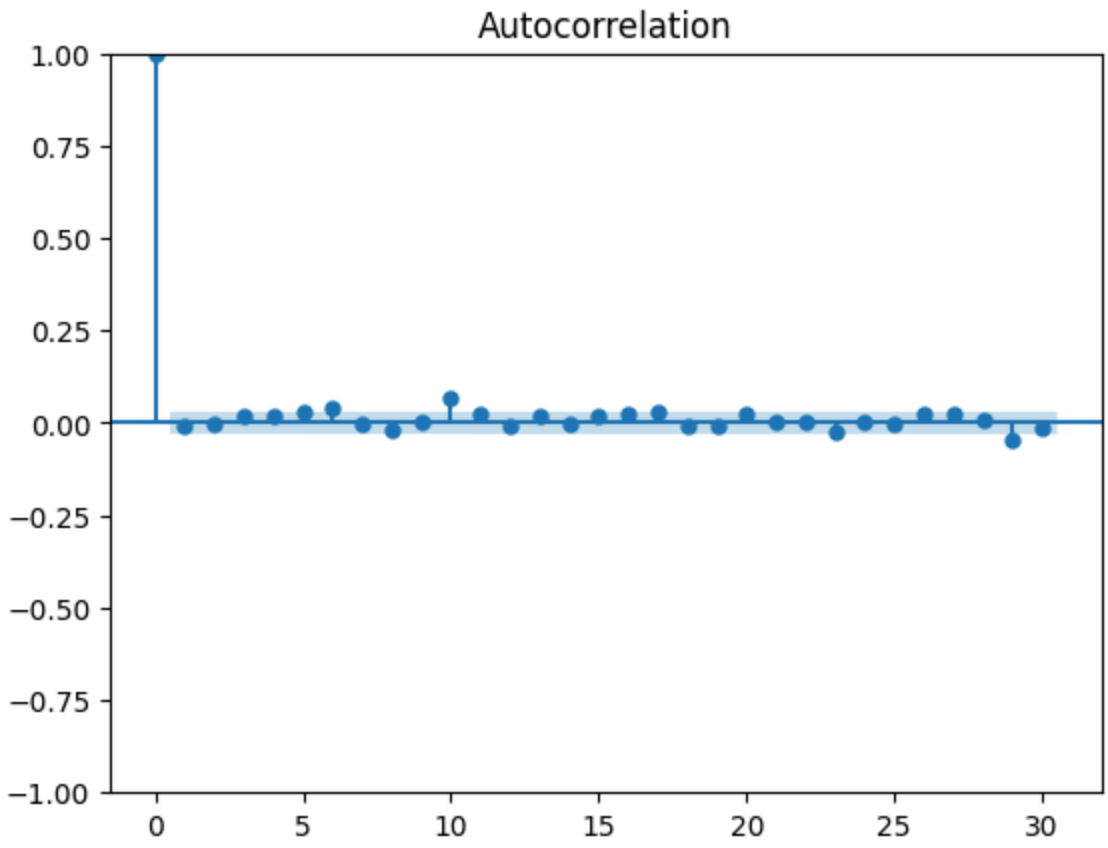
Diagnosis: After extracting the features, diagnostic plots can be used to assess how well the features capture the underlying patterns in the data. The rolling mean and volatility, for instance, can help identify trends and periods of high uncertainty in Bitcoin and stocks price behaviour (Tse, 2002). To diagnose the model's effectiveness, there is a need to use several diagnostic tools and plots.

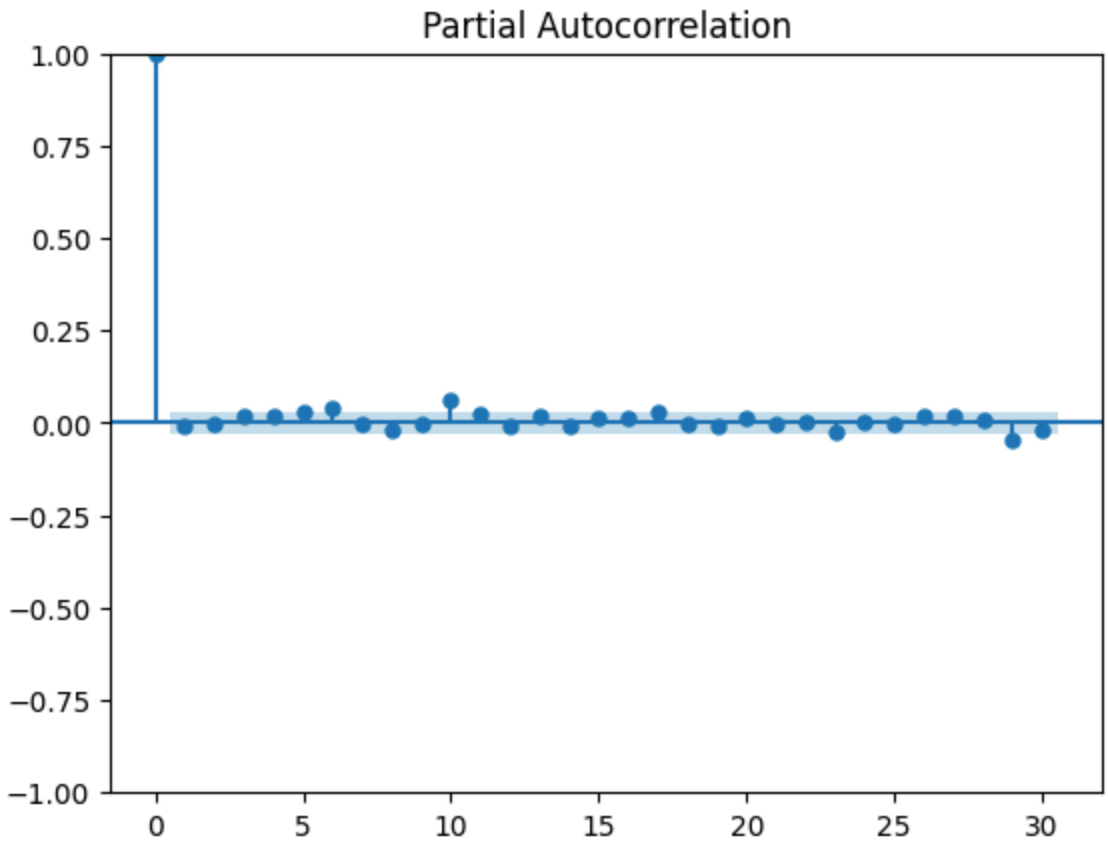
Residual Plot (for Volatility Model)



Interpretation: The residual plot shows the residuals of the log return model. If the residuals are randomly distributed around zero, it indicates that the model is a good fit. However, any patterns in the residuals could indicate model misspecification.

Autocorrelation Plot (ACF/PACF)


Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25
30



Interpretation: The Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots provide insight into the correlation of the time series with its own lagged values. This can help identify if there is any serial dependence in the data.

Damage: Overfitting can be a concern when using too many features. It is important to analyze the correlation between features and avoid using redundant variables in the model (James et al., 2013).

Directions: To improve model performance, feature selection techniques such as Principal Component Analysis (PCA) or feature regularization methods like Lasso or Ridge regression can help reduce the dimensionality and eliminate multicollinearity (Tibshirani, 1996).

Deployment: The extracted features can be utilized in various predictive modeling techniques such as linear regression, machine learning algorithms (e.g., Random Forest or Gradient Boosting), or deep learning models like Recurrent Neural Networks (RNNs). These models can be used to forecast Bitcoin and stocks prices or assess investment strategies (Chakraborty & Johnson, 2011).

Conclusion: By combining feature extraction with advanced predictive models, Bitcoin and stocks prices forecasting can become more effective and also help in understanding market conditions, thereby allowing investors to make more informed decisions.

FootNote

Advantage of using AI:

Needing less time to collect information and to evaluate it for accuracy as compared with using bibliographical research.

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